

Demonstration of Proposed Features

Date	04 july 2026
Team ID	
Project Name	Credit Card Approval Prediction
Maximum Marks	1 Mark

Demonstration of Proposed Features:

This document captures all the features that were proposed during the project planning phase and tracks whether each feature was successfully implemented and demonstrated. It serves as evidence of the team's ability to deliver on the proposed Credit Card Approval Prediction System solution.

S.No	Feature Name	Description	Status(Implemented / Partial / Pending)	Demonstrated(Yes / No)	Remarks
1	Data Preprocessing & Feature Engineering	Handled missing values, removed duplicates, encoded categorical variables, and converted payment status into binary labels.	Implemented	Yes	Clean dataset produced for reliable model training.
2	Exploratory Data Analysis & Visualization	Analyzed applicant features (Gender, Income Type, Annual Income, Employment Duration, Education) using count and distribution plots.	Implemented	Yes	Key approval/rejection trends identified via Matplotlib and Seaborn.
3	Multi-Model Classification Training	Trained Logistic Regression, Decision Tree, Random Forest, and XGBoost classifiers on the processed dataset.	Implemented	Yes	All four models trained and compared successfully.
4	Model Evaluation & Best Model Selection	Evaluated models using accuracy score, confusion matrix, and classification report; selected and saved the best-performing model.	Implemented	Yes	Best model exported for integration into the web application.
5	Flask Web Application for Real-Time Prediction	Built an interactive Flask app with a home page, prediction interface, user input forms, and result display.	Implemented	Yes	Users can submit applicant details and get instant approval/rejection predictions.
6	IBM Watson Machine Learning Cloud Deployment	Deployed the trained model on IBM Watson ML for scalable, real-time predictions accessible via the web UI.	Implemented	Yes	Solution successfully hosted on the cloud and accessible end-to-end.

Feature Implementation Summary:

Total Features Proposed	6
Total Features Implemented	6
Total Features Demonstrated	6
Overall Implementation Rate (%)	100%

